

**VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.10)**

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1. EXECUTIVE SUMMARY

This report documents validation of the Shielding Attenuation Simulator V1.10. Version 1.10 adds a reusable engineering visualization layer above the previously validated V1.09 constraint-based material optimizer. It converts optimization results into structured plot data, classifies thickness mass Pareto tradeoffs, samples detector response versus shield thickness, renders four portfolio-quality engineering figures, and exports reproducible PNG and SVG artifacts.

The Beer-Lambert attenuation model, NIST XCOM interpolation, inverse-square point-source model, isotope source calculations, G-P exposure buildup equation, minimum thickness solver, material comparison, engineering constraints, and optimization ranking are unchanged from V1.09. Therefore, V1.10 does not require a second full derivation or independent revalidation of those physics calculations. The V1.09 validation report remains the controlling validation record for the underlying attenuation and optimization mathematics.

A focused V1.10 test program is still necessary because the release introduces new software and numerical behavior beyond drawing static pictures. New logic includes candidate-to-plot data conversion, preservation of failed and rejected states, Pareto dominance classification, exact insertion of optimized thicknesses into sampled curves, manual and isotope response curve evaluation, handling of unavailable buildup values, constraint feasibility classification, and noninteractive figure generation. These areas were validated without duplicating the long physics test tables from V1.09.

The complete validation runner produced 273 PASS assertions and ended with "All validation tests passed." Of these, 58 assertions directly exercise V1.10 behavior. The reproducible figure generation script executed successfully, selected lead at 7.08821 cm for the documented constrained Cs-137 case, and produced four PNG and four SVG files. All eight artifacts were nonempty; PNG files decoded successfully and SVG files parsed as valid XML.

2. SCOPE AND VALIDATION METHOD

2.1 Previously Validated Calculations

V1.10 consumes OptimizedDesignResult objects produced by the V1.09 optimizer. The required thickness, mass per area, relative cost index, eligibility status, rejection reasons, selected candidate, and buildup state originate from previously validated calculation pathways. The visualization subsystem does not modify those values or rerun optimization inside the plotting functions.

2.2 New V1.10 Components

- `plot_models.py` defines validated, plot-ready data structures for material points, Pareto points, feasibility rows, and response curves.
- `plot_data.py` converts V1.09 optimization results into generic plot data and performs Pareto and constraint-feasibility classification independently of Matplotlib.
- `response_curve.py` validates thickness ranges, creates ordered samples, evaluates manual or isotope source response at each thickness, preserves narrow-beam and G-P results separately, and inserts the exact selected thickness.
- `optimization_plots.py` and `response_plots.py` render the four finalized engineering figures without changing the underlying numerical data.
- `plotting_utils.py` centralizes typography, colors, legends, text formatting, and file export.
- `examples/generate_v110_figures.py` reproduces the official Cs-137 case using a noninteractive Matplotlib backend and saves PNG and SVG outputs.

2.3 Validation Philosophy

Plot validation is based on data integrity and deterministic file behavior rather than pixel-by-pixel image comparison. Exact pixel comparison would be fragile across operating systems, Matplotlib versions, and font installations and would not provide meaningful scientific assurance. Instead, validation confirms that the plotted values and statuses match the optimizer, the Pareto front is mathematically correct, response curves follow the existing calculation pathways, invalid requests are rejected, and expected output files are generated successfully. Final visual layout was additionally reviewed for clipping, overlap, labeling, units, and figure-level context.

3. TEST CASE RESULTS

3.1 Automated Validation Summary

Validation Area	Acceptance Criterion	Result
Plot-data integrity	Candidate count, metrics, selected material, statuses, warnings, and rejection reasons preserved	PASS
Failed-candidate handling	No fabricated thickness, mass, cost, or feasibility values; failure reason preserved	PASS
Pareto classification	Lead and tungsten form the feasible thickness mass Pareto front; points sorted deterministically	PASS
Thickness sampling	Endpoints preserved, samples strictly increasing, exact required thickness inserted	PASS
Response calculations	Zero-thickness values, monotonic narrow-beam response, target conversion, and target intersection verified	PASS
Buildup availability	Unsupported and out-of-range buildup remains unavailable and generates warnings; no silent substitution	PASS
Constraint feasibility	PASS, FAIL, INACTIVE, and UNAVAILABLE states verified, including equality at active limits	PASS
Input rejection	Thickness beyond detector distance and insufficient sample count rejected	PASS
Regression suite	All earlier attenuation, buildup, source, minimum thickness, comparison, and optimization tests retained	PASS

The current runner contains 273 PASS assertions. The V1.09 report documented 215 PASS assertions; V1.10 adds 58 focused assertions while retaining the complete earlier regression suite.

3.2 Plot Data and Status Preservation

The plot data layer was tested independently of rendering. The constrained Cs-137 optimization result was converted into nine MaterialPlotPoint objects, and the validation runner confirmed that the selected lead candidate, rejected statuses, multiple rejection reasons, buildup displays, and failed candidate metadata were preserved. Failed candidates must not receive artificial numerical coordinates; unavailable constraint checks remain explicitly unavailable.

3.3 Pareto Classification

A candidate is Pareto optimal in thickness and mass when no other feasible candidate is both no thicker and no heavier, with at least one strict improvement. For the official constrained case, lead and tungsten are the only eligible designs and neither dominates the other: tungsten is

thinner, while lead has lower mass per area. Both are therefore members of the feasible Pareto front. Rejected designs remain visible for engineering context but do not define the feasible front.

3.4 Response Curve Sampling

Response curve validation confirmed that the zero thickness point equals the unshielded source response, narrow-beam response decreases monotonically with increasing thickness, all stored values remain nonnegative, and the exact optimized thickness is inserted even when it does not coincide with the uniform sampling grid. Manual monoenergetic and isotope sources were tested through their respective existing calculation pathways.

For lead in the official Cs-137 case, the exact selected thickness is 7.088213 cm. At this point the narrow-beam uncollided photon flux is approximately 33.1023 photons/cm²/s, while the G-P exposure-buildup target-comparison response is 99.99994 photons/cm²/s, reproducing the 100 photons/cm²/s design target within the minimum thickness solver tolerance.

3.5 Artifact Generation Verification

Figure	Format	File Size	Result
Constraint feasibility	PNG	326,854 bytes	PASS
Constraint feasibility	SVG	109,663 bytes	PASS
Lead response vs. thickness	PNG	519,814 bytes	PASS
Lead response vs. thickness	SVG	149,828 bytes	PASS
Material design comparison	PNG	557,347 bytes	PASS
Material design comparison	SVG	169,344 bytes	PASS
Thickness mass tradeoff	PNG	390,039 bytes	PASS
Thickness mass tradeoff	SVG	130,582 bytes	PASS

The official generation script uses the Agg noninteractive backend, creates the output directory, saves each figure in both formats, closes each Matplotlib figure, and prints a deterministic summary. PNG dimensions ranged from 3003 x 2046 to 4457 x 2145 pixels. Each PNG decoded successfully and each SVG parsed as valid XML. File sizes are recorded for traceability only. Successful decoding or parsing and nonzero output size are the acceptance criteria; exact byte counts are not regression thresholds.

4. OFFICIAL V1.10 ENGINEERING CASE

4.1 Input Scenario

Parameter	Value
Source	Cs-137
Activity	1.00 Ci (3.70×10^{10} Bq)
Detector distance	100 cm
Design target	Target-comparison response ≤ 100 photons/cm ² /s
Calculation search limit	100 cm
Maximum design thickness	20 cm
Maximum mass per area	120 g/cm ²
Optimization objective	Minimum mass per area
Buildup setting	G-P exposure buildup requested
Candidate materials	Nine materials in the Python material library

4.2 Selected Design and Engineering Interpretation

Lead was selected at 7.08821 cm with a mass per area of 80.3803 g/cm² and a relative cost index per area of 241.141. Tungsten was also eligible and required only 4.90817 cm, but its higher density produced a greater mass per area of 94.5020 g/cm². All other materials violated at least one active engineering constraint.

4.3 Candidate Results

The following table reproduces the numerical values shown in the material comparison, feasibility, and Pareto figures. Status reflects the two active engineering constraints before application of the minimum mass objective.

Material	Thickness (cm)	Mass/Area (g/cm ²)	Rel. Cost/Area	Status
Lead	7.08821	80.3803	241.141	ELIGIBLE / SELECTED
Tungsten	4.90817	94.5020	2362.55	ELIGIBLE
Copper	17.2589	154.640	1237.12	REJECTED - mass
Tin	18.6007	135.971	951.799	REJECTED - mass
Concrete (Barite, Type BA)	28.7625	100.669	161.070	REJECTED - thickness
Aluminum	58.3158	157.453	629.810	REJECTED - thickness and mass
Concrete (Ordinary)	62.0647	148.955	148.955	REJECTED - thickness and mass
Water	91.3324	91.3324	22.8331	REJECTED - thickness
Polyethylene	94.6039	88.9277	177.855	REJECTED - thickness

Buildup aware thicknesses are used for lead, tungsten, aluminum, ordinary concrete, copper, and tin. Barite concrete and polyethylene use documented narrow-beam fallback because G-P coefficients are unavailable. Water also uses fallback because its buildup aware target is not reached within the calculation range.

4.4 Figure Interpretation

- The material comparison figure shows that thickness efficiency, mass efficiency, and simplified cost efficiency produce different rankings; no single material is best in every metric.
- The feasibility matrix explains rejection directly: copper and tin satisfy the thickness limit but fail the mass limit, while water and polyethylene satisfy the mass limit but fail the thickness limit.
- The Pareto figure separates feasible tradeoffs from infeasible designs. Tungsten cannot dominate lead because it is heavier, and lead cannot dominate tungsten because it is thicker.
- The response curve shows why the selected thickness is greater than a narrow-beam-only estimate. The G-P exposure-buildup response proxy reaches the design target at the selected point while the uncollided component is lower.

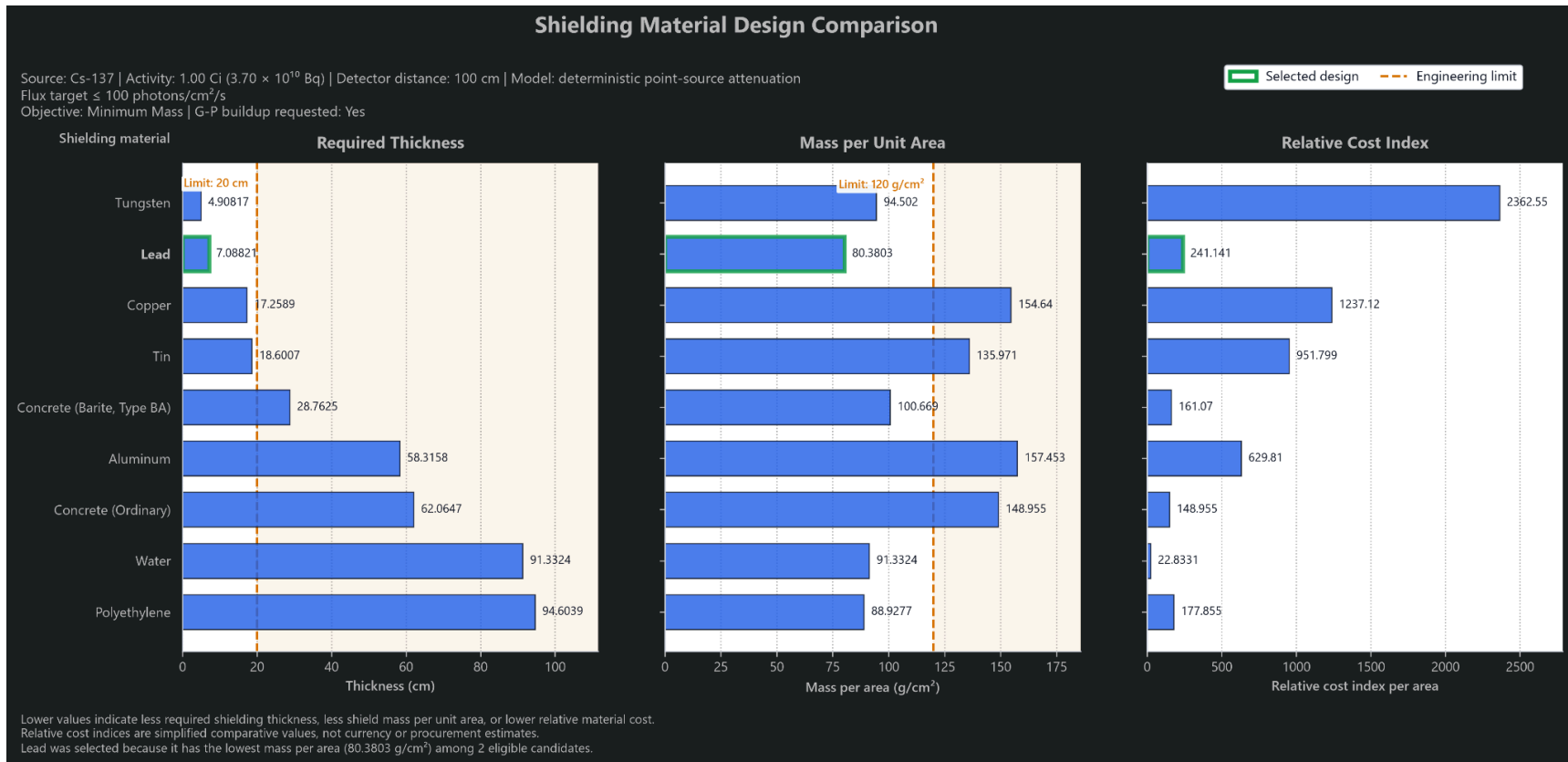


Figure 1. Required thickness, mass per unit area, and relative cost index for the shared Cs-137 design target. The selected lead design is outlined in green.

Engineering Constraint Feasibility by Material

Source: Cs-137 | Activity: 1.00 Ci (3.70×10^{10} Bq) | Detector distance: 100 cm | Model: deterministic point-source attenuation
 Flux target ≤ 100 photons/cm²/s
 Objective: Minimum Mass | G-P buildup requested: Yes

Material	Thickness ≤ 20 cm	Mass per area ≤ 120 g/cm ²	Overall result
Lead	PASS	PASS	SELECTED
Tungsten	PASS	PASS	ELIGIBLE
Aluminum	FAIL	FAIL	REJECTED
Concrete (Barite, Type BA)	FAIL	PASS	REJECTED
Concrete (Ordinary)	FAIL	FAIL	REJECTED
Copper	PASS	FAIL	REJECTED
Polyethylene	FAIL	PASS	REJECTED
Tin	PASS	FAIL	REJECTED
Water	FAIL	PASS	REJECTED

Each constraint is evaluated independently. A material is eligible only when it passes every active engineering constraint. Lead was selected because it has the lowest mass per area (80.3803 g/cm²) among 2 eligible candidates.

Figure 2. Independent evaluation of the 20 cm thickness limit and 120 g/cm² mass per area limit. Lead and tungsten satisfy both active constraints.

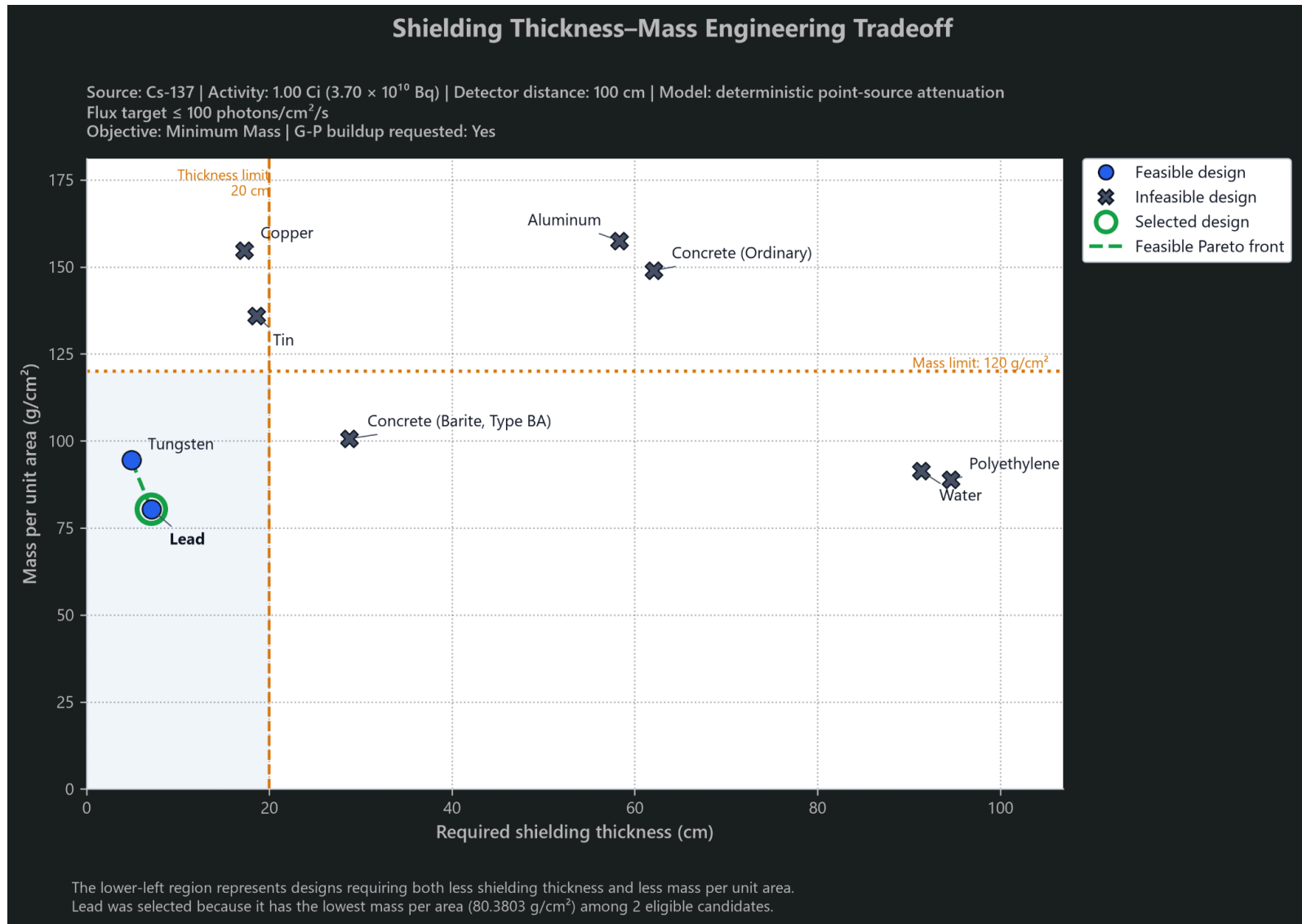


Figure 3. Thickness mass engineering tradeoff. Lead and tungsten form the feasible Pareto front; lead is selected under the minimum mass objective.

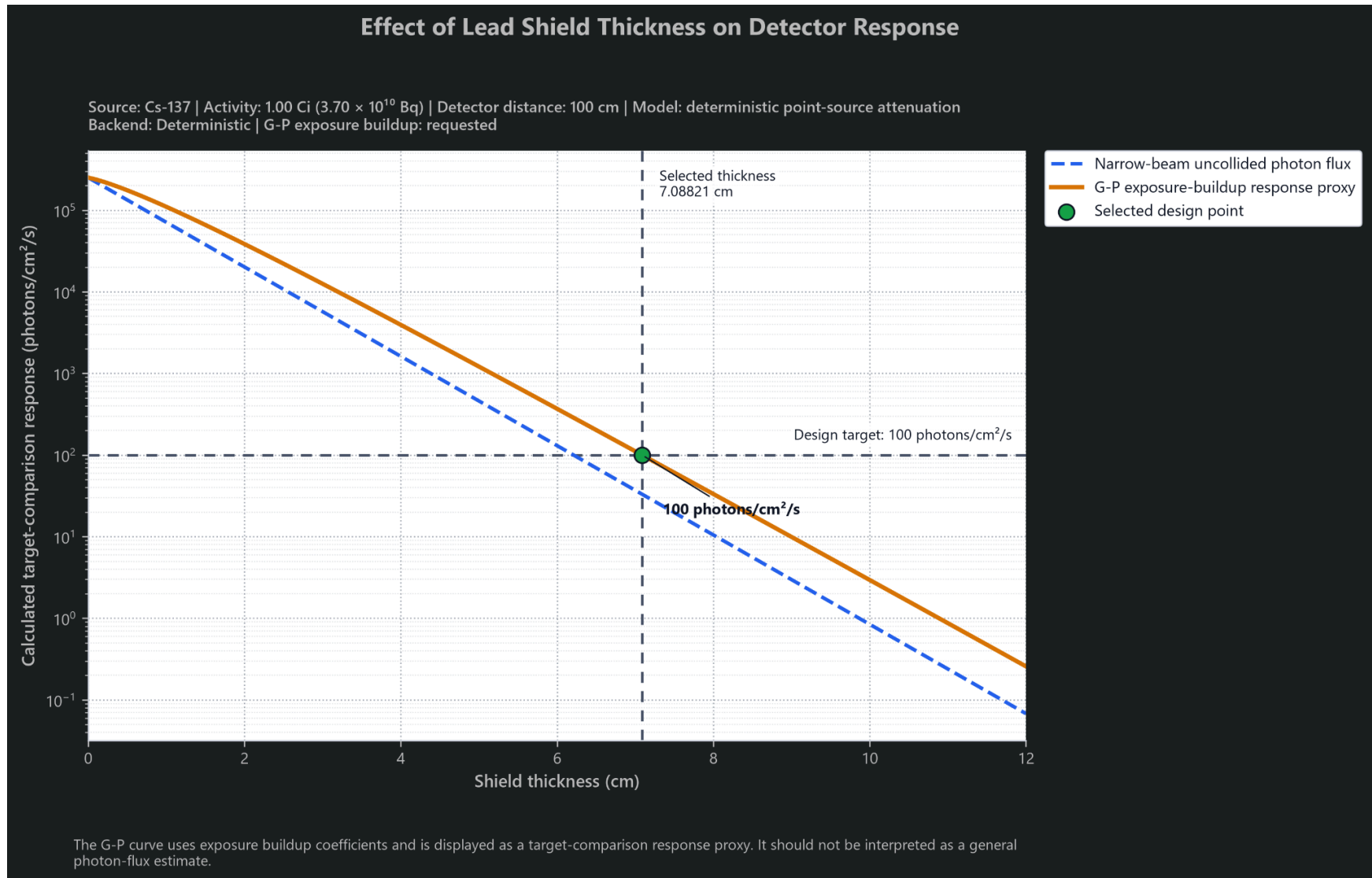


Figure 4. Lead detector response versus shielding thickness. The G-P curve is an exposure-buildup target comparison response proxy, not a general photon flux estimate.

5. ASSUMPTIONS AND LIMITATIONS

1. Previously Validated Physics Reused: V1.10 validates visualization data, Pareto logic, response sampling, and artifact generation rather than independently revalidating the V1.09 attenuation and optimization equations.
2. No Pixel-Perfect Regression Test: Figure appearance is reviewed visually and output artifacts are checked structurally. Exact pixel matching is intentionally avoided because fonts and rendering can vary by platform.
3. Deterministic Point-Source Geometry: The official case retains the existing isotropic point-source and one-dimensional source-to-detector shielding geometry.
4. Single-Material Designs: The figures compare homogeneous single-material shields. Multilayer transport and two-layer optimization remain outside V1.10.
5. Mass per Area, Not Total Mass: Without a complete shield geometry, mass is represented as density multiplied by thickness in g/cm^2 .
6. Relative Cost Index, Not Currency: Cost values are simplified, user-editable comparative indices and are not market prices or procurement estimates.
7. Exposure Buildup Coefficients: The implemented G-P tables represent exposure buildup. The plotted buildup corrected curve is therefore labeled a target-comparison response proxy and must not be interpreted as a general photon-flux estimate.
8. Mixed Buildup Availability: Lead, aluminum, ordinary concrete, tungsten, copper, and tin use supported G-P buildup in the official comparison. Barite concrete and polyethylene use documented narrow-beam fallback; water also falls back because the buildup aware target is unavailable within the calculation range.
9. 40-MFP Buildup Limit: The response sampler stops the buildup series when the validated G-P range is exceeded and does not extrapolate or silently substitute narrow-beam values.
10. Selected Isotope Lines: Cs-137 calculations use the implemented selected photon-line library rather than a complete evaluated decay spectrum.
11. No Dose or Regulatory Claim: V1.10 does not calculate effective dose, certify shielding performance, or replace detailed transport analysis and regulatory review.
12. No Monte Carlo Benchmark Yet: OpenMC comparison is reserved for later versions and is not part of this validation claim.

6. CONCLUSION

The Shielding Attenuation Simulator V1.10 successfully adds a reusable and reproducible engineering visualization subsystem while preserving all previously validated attenuation and optimization behavior. The release converts V1.09 results into validated plot data objects, identifies feasible Pareto tradeoffs, samples manual and isotope response curves, handles unavailable buildup data transparently, and produces four report-quality figures in PNG and SVG formats.

The report is intentionally shorter than the V1.09 validation report because V1.10 does not introduce a new attenuation equation or optimization objective. It does, however, introduce enough numerical and software logic to require focused validation. The 58 new V1.10 assertions confirm plot data integrity, Pareto classification, response sampling, constraint feasibility states, buildup availability behavior, and invalid input rejection. Together with the retained regression suite, the complete runner produced 273 PASS assertions and no failures.

The official 1 Ci Cs-137 case reproduced the V1.09 engineering result: lead is selected at 7.08821 cm because it has the lowest mass per area among the two candidates satisfying both active constraints. The finalized figures make that result and its tradeoffs directly understandable to technical reviewers and provide a stable visualization foundation for future shared scenario models, response quantities, OpenMC benchmarks, multilayer design, and the planned graphical user interface.

References

- [1] Y. Harima, Y. Sakamoto, S. Tanaka, and M. Kawai, “Validity of the Geometric-Progression Formula in Approximating Gamma-Ray Buildup Factors,” *Nuclear Science and Engineering*, vol. 94, no. 1, pp. 24–35, Sep. 1986. doi:10.13182/nse86-a17113
- [2] D. K. Trubey, “New gamma-ray buildup factor data for point kernel calculations: ANS-6.4.3 standard reference data,” IAEA, <https://inis.iaea.org/records/0arzw-ez976/files/20014493.pdf> (accessed Jul. 10, 2026).
- [3] Angstrom Sciences, “Density of elements chart – Angstrom Sciences elements density table,” Angstrom Sciences, <https://angstromsciences.com/density-elements-chart> (accessed Mar. 26, 2026).
- [4] J. Colledge, M. Wilczek, and C. Raymond, “Inverse square law | radiology reference article,” Radiopaedia, <https://radiopaedia.org/articles/inverse-square-law?lang=us> (accessed Mar. 26, 2026).
- [5] “Shielding Radiation Alphas, Betas, Gammas and Neutrons,” U.S. NRC, <https://www.nrc.gov/docs/ML1122/ML11229A721.pdf> (accessed Mar. 26, 2026).
- [6] M. J. Berger et al., “XCOM: Photon Cross Sections Database,” NIST, <https://dx.doi.org/10.18434/T48G6X> (accessed Mar. 26, 2026).
- [7] National Nuclear Data Center, “NuDat 3 Glossary,” Brookhaven National Laboratory, <https://www.nndc.bnl.gov/nudat3/guide/glossary.html> (accessed Jul. 11, 2026).
- [8] International Atomic Energy Agency, “LiveChart of the Nuclides,” IAEA Nuclear Data Section, <https://nds.iaea.org/relnsd/vcharthtml/VChartHTML.html> (accessed Jul. 11, 2026).
- [9] J. D. Hunter, “Matplotlib: A 2D Graphics Environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007. doi:10.1109/MCSE.2007.55 (accessed Jul. 20, 2026).